Cancer Resistance in Elephants: A Multi-Species Analysis of Neoplasia Prevalence

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Abstract

This study investigates cancer prevalence in elephants. Elephant species, including *Elephas maximus, Loxodonta africana, and Loxodonta cyclotis,* have historically exhibited low malignancy rates, potentially a result of multiple cancer-protective genes, such as LIF6 (Leukemia inhibitory factor 6) and TP53 (tumor protein 53). However, research on prevalence is limited by a lack of integration across independent datasets. To address this gap, this study builds on past research through detailed analysis of necropsy data compiled from multiple sources, including captive elephant necropsy records from zoos, reproductive tract neoplasia reports, and multi-species cancer datasets, to comprehensively assess neoplasia and malignancy prevalence across elephant species. Findings across datasets indicate consistent trends of both benign and malignant neoplasm occurrence. While species-dependent variation exists between *L. africana* and *E. maximus*, benign neoplasia rates range from as low as 1.05% up to 8.7%, which is lower than the mean neoplasia rate of 18.4% observed among other large mammals. Malignant occurrences are low in *L. africana* and *E. maximus* and range from 1.4% to 3%. Reproductive tract neoplasms, particularly leiomyomas are more frequently observed in *E. maximus*, which suggests the possibility of a species- and sex- specific susceptibility. These results reinforce past assessments that have revealed low malignancy prevalence among elephants and identify a species-specific difference, with overall neoplasms consistently occurring at lower rates in *L. africana* than. *E. maximus*.

Neoplasia Prevalence

Cumulative Neoplasia and Malignancy Results		
Classification	**Count**	**Percentage**
Benign	153	5.69%
Malignancy	52	1.93%
Reproductive Benign or Malignancy	98	3.64%
Total Records	**2687**	

Table 1. Cumulative Neoplasia and Malignancy Results

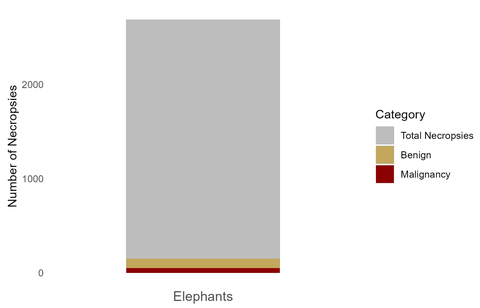

Figure 1. Out of 2,687 total necropsy records, 153 cases involved benign neoplasias and 52 involved malignant neoplasias. Out of 2,687 total necropsy records, 153 cases involved benign neoplasias and 52 involved malignant neoplasias.

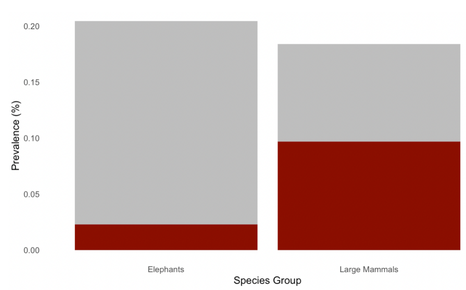

Figure 2. Comparison of Benign and Malignant Neoplasia Prevalence in Elephant Species Versus Other Large Mammals. Elephants exhibited a slightly higher prevalence of benign neoplasia compared to other large mammals (20.45% vs. 18.42%), while the prevalence of malignancy was markedly lower in elephants (2.27% vs. 9.69%).

Neoplasia Demographics

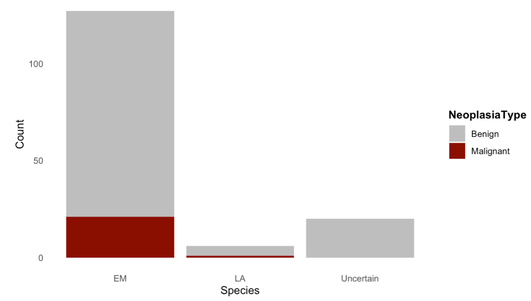

Figure 3. Neoplasia prevalence by species. *E. maximus* exhibited the highest prevalence of both benign and malignant neoplasias, with 127 total neoplasia cases and 44 malignancies out of 1,458 records. *L. africana* had 6 neoplasia cases and 8 malignancies out of 567 records. *L. cyclotis* had no reported cases from 15 records.

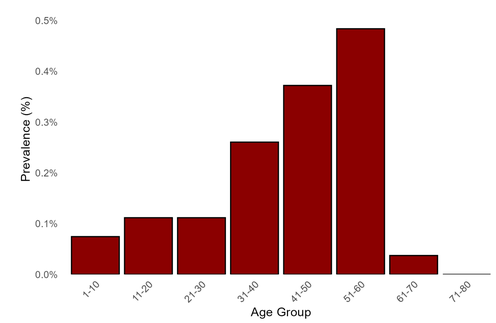

Figure 4. Malignancy prevalence by age group. The 51-60 group experienced the highest prevalence with 13 cases. There is a notable lack of malignancy occurence in ages over 60, but this may be due to a smaller sample size. Ages 1-10 had 1 case, 11-20 and 21-30 had 3 cases, 31-40 had 7 cases, and 41-50 had 10 cases.

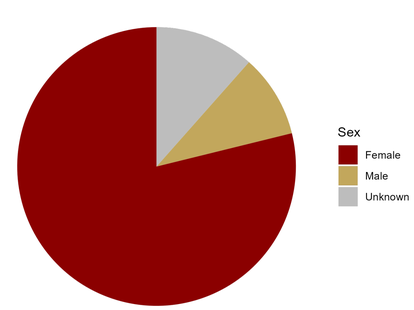

Figure 5 Sex breakdown among malignancy cases. Female individuals accounted for 78.85% of malignancy cases, males for 9.62%, and 11.54% were classified as unknown sex.

Reproductive Neoplasia Incidence

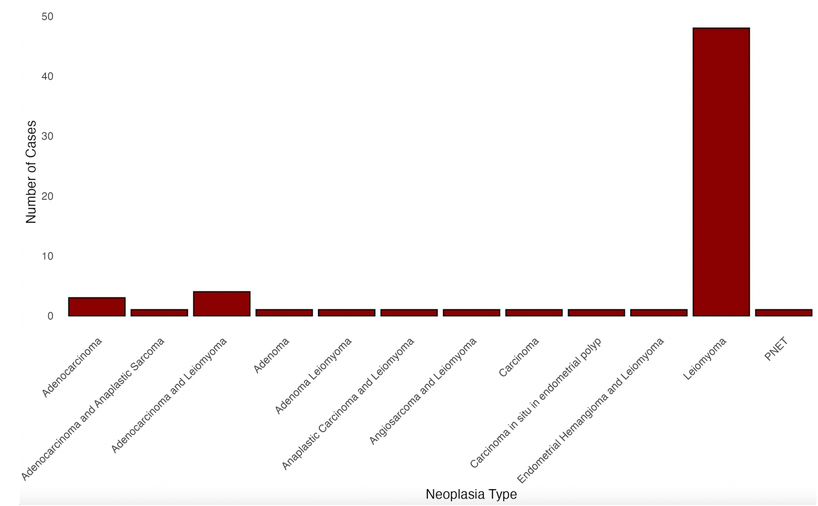

Figure 6. Reproductive neoplasias are the most commonly observed category of neoplastic disease in elephants. Leiomyomas (benign) represent the most frequently diagnosed individual tumor type. The second most frequent diagnosis involves co-occurring leiomyomas and adenocarcinomas (malignant), followed by adenocarcinomas alone.

Conclusion and Future Directions

Elephants have long served as the quintessential demonstration of Peto's Paradox; the theoretical concept that suggests a disconnect between body mass and expected cancer prevalence. By utilizing multiple independent sets of necropsy data, featuring three separate elephant species, this study demonstrated a consistent pattern of exceptionally low malignancy prevalence when compared to other mammals of similar body mass. Distribution of malignancies across age groups saw higher occurence with increased age, peaking at the 51-60 age group. There is a drastic decrease in malignancy prevalence in age groups 61-70 and 71-80, which may be attributed to the relatively small sample sizes of these groups. The data further suggested a sex-specific vulnerability due to the high occurrence of female-reproductive tract neoplasms. *E. maximus* appeared to demonstrate a higher malignancy prevalence when compared to *L. africana*, suggesting species-specific vulnerability, but this relationship was limited by the relatively small sample size of *L. africana* elephants. Both the species- and sex-specific vulnerabilities may serve as future points of research. The near absence of *L. cyclotis* in datasets also warrants investigation to further clarify interspecies differences and potential susceptibilities.

Acknowledgements

We thank Prof. Carlo Maley for the opportunity to be associates of The Arizona Cancer Evolution Center, as well as Walker Mellon, Gissel Marquez Alcaraz, and Harley Richter for their ongoing leadership in the ACE Scholars program.

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